

# EduFin SQL Business Challenge

Phase 4 — Thursday Submission

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## STEP 4A — Open Metric Design: "Partner Health Score"

```
WITH inst_summary AS (
    SELECT
        i.institution_id,
        i.institution_name,
        i.institution_type,
        COUNT(*) AS loan_count,
        COUNT(DISTINCT l.customer_id) AS customers,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name, i.institution_type
),
multiplier_data AS (
    SELECT
        *,
        (default_count * 100.0 / loan_count) AS default_rate,
        CASE
            WHEN loan_count >= 200 THEN 1.5
            WHEN loan_count >= 50 THEN 1.0
            ELSE 0.5
        END AS volume_multiplier
    FROM inst_summary
)
SELECT
    institution_id AS `Institution Id`,
    institution_name AS `Institution Name`,
    institution_type AS `Institution Type`,
    customers AS `Customers`,
    loan_count AS `Loans`,
    default_count AS `Default Count`,
    CONCAT(ROUND(default_rate, 2), '%') AS `Default Rate (%)`,
    ROUND((1 - default_rate / 100) * volume_multiplier * 100, 2) AS `Health Score`
FROM multiplier_data
ORDER BY `Health Score` DESC;
```

## STEP 4B — Default Rate Per Institution

```
WITH default_stats AS (
    SELECT
        i.institution_id,
        i.institution_name,
        COUNT(*) AS loan_count,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
    FROM institutions i
    INNER JOIN loans l
        ON i.institution_id = l.institution_id
    GROUP BY i.institution_id, i.institution_name
    HAVING COUNT(l.loan_id) >= 50
)
SELECT
```

```

institution_id AS `Institution ID`,
institution_name AS `Institution Name`,
loan_count AS `Loans`,
default_count AS `Default Count`,
CONCAT(default_rate, '%') AS `Default Rate (%)`
FROM default_stats
ORDER BY default_rate DESC, default_count DESC;

```

| Institution ID | Institution Name                 | Loans | Default Count | Default Rate (%) |
|----------------|----------------------------------|-------|---------------|------------------|
| 29             | State Commerce Institute         | 10005 | 2088          | 20.87%           |
| 33             | Central Engineering Institute    | 9878  | 2050          | 20.75%           |
| 34             | Modern Management Institute      | 10165 | 2091          | 20.57%           |
| 31             | Regional Commerce University     | 9915  | 2039          | 20.56%           |
| 30             | National Engineering Institute   | 9889  | 2017          | 20.40%           |
| 32             | National Commerce Institute      | 9947  | 2028          | 20.39%           |
| 50             | Government Medical College       | 9780  | 1967          | 20.11%           |
| 28             | Regional Medical University      | 10134 | 1737          | 17.14%           |
| 25             | Central Commerce College         | 10009 | 1688          | 16.86%           |
| 26             | State Arts & Science University  | 10108 | 1687          | 16.69%           |
| 24             | Modern Law College               | 10151 | 1682          | 16.57%           |
| 27             | Central Management College       | 9984  | 1641          | 16.44%           |
| 23             | Advanced Technology University   | 10121 | 1652          | 16.32%           |
| 36             | Indian Arts & Science University | 9897  | 1324          | 13.38%           |
| 49             | Advanced Commerce University     | 9926  | 1311          | 13.21%           |
| 8              | State Engineering Institute      | 9860  | 1300          | 13.18%           |
| 13             | State Medical University         | 10259 | 1330          | 12.96%           |
| 38             | Central Management College       | 10039 | 1293          | 12.88%           |
| 12             | Central Commerce University      | 10082 | 1291          | 12.80%           |
| 21             | Central Law University           | 9956  | 1273          | 12.79%           |
| 46             | Regional Law College             | 9886  | 1254          | 12.68%           |
| 7              | Indian Law University            | 10024 | 1270          | 12.67%           |
| 48             | Central Commerce University      | 9916  | 1254          | 12.65%           |
| 4              | Advanced Arts & Science College  | 10166 | 1282          | 12.61%           |
| 2              | Central Law University           | 10107 | 1274          | 12.61%           |

## STEP 4C — Financial Loss Attribution

```

WITH financial_exposure AS (
  SELECT
    i.institution_id,
    i.institution_name,
    SUM(l.loan_amount) / 100000000.0 AS loan_amount,
    ROUND(SUM(CASE WHEN l.loan_status = 'Overdue' THEN l.loan_amount * 1.0 ELSE 0
END) / 100000000.0, 2) AS overdue_amount,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 100000000.0, 2) AS default_amount
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  CONCAT(ROUND(loan_amount, 2), ' Cr') AS `Loan Amount`,
  CONCAT(ROUND(overdue_amount, 2), ' Cr') AS `Overdue Amount`,
  CONCAT(ROUND(default_amount, 2), ' Cr') AS `Default Amount`,
  CONCAT(ROUND(overdue_amount + default_amount, 2), ' Cr') AS `At-Risk Exposure`
FROM financial_exposure
ORDER BY default_amount DESC;

```

|    | Institution ID | Institution Name                 | Loan Amount | Overdue Amount | Default Amount | At-Risk Exposure |
|----|----------------|----------------------------------|-------------|----------------|----------------|------------------|
| 34 |                | Modern Management Institute      | 461.35 Cr   | 34.62 Cr       | 88.54 Cr       | 123.16 Cr        |
| 31 |                | Regional Commerce University     | 448.56 Cr   | 32.84 Cr       | 85.99 Cr       | 118.83 Cr        |
| 30 |                | National Engineering Institute   | 451.53 Cr   | 36.91 Cr       | 85.15 Cr       | 122.06 Cr        |
| 29 |                | State Commerce Institute         | 446.11 Cr   | 37.21 Cr       | 83.96 Cr       | 121.17 Cr        |
| 33 |                | Central Engineering Institute    | 447.13 Cr   | 34.52 Cr       | 83.5 Cr        | 118.02 Cr        |
| 32 |                | National Commerce Institute      | 454.88 Cr   | 36.59 Cr       | 82.81 Cr       | 119.4 Cr         |
| 50 |                | Government Medical College       | 437.3 Cr    | 33.4 Cr        | 79.94 Cr       | 113.34 Cr        |
| 28 |                | Regional Medical University      | 454.79 Cr   | 34.74 Cr       | 68.86 Cr       | 103.6 Cr         |
| 26 |                | State Arts & Science University  | 456.33 Cr   | 38.01 Cr       | 68.38 Cr       | 106.39 Cr        |
| 25 |                | Central Commerce College         | 448.52 Cr   | 37.63 Cr       | 66.67 Cr       | 104.3 Cr         |
| 24 |                | Modern Law College               | 450.19 Cr   | 34.88 Cr       | 64.85 Cr       | 99.73 Cr         |
| 27 |                | Central Management College       | 443.59 Cr   | 36.46 Cr       | 64.64 Cr       | 101.1 Cr         |
| 23 |                | Advanced Technology University   | 446.03 Cr   | 37.48 Cr       | 63.76 Cr       | 101.24 Cr        |
| 13 |                | State Medical University         | 459.15 Cr   | 39.49 Cr       | 51.87 Cr       | 91.36 Cr         |
| 12 |                | Central Commerce University      | 457.47 Cr   | 35.51 Cr       | 51.42 Cr       | 86.93 Cr         |
| 36 |                | Indian Arts & Science University | 448.59 Cr   | 36.95 Cr       | 50.59 Cr       | 87.54 Cr         |
| 38 |                | Central Management College       | 457.73 Cr   | 38.54 Cr       | 50.51 Cr       | 89.05 Cr         |
| 7  |                | Indian Law University            | 451.27 Cr   | 35.6 Cr        | 49.63 Cr       | 85.23 Cr         |
| 4  |                | Advanced Arts & Science College  | 451.9 Cr    | 34.66 Cr       | 49.19 Cr       | 83.85 Cr         |
| 21 |                | Central Law University           | 453.34 Cr   | 33.36 Cr       | 49.09 Cr       | 82.45 Cr         |
| 2  |                | Central Law University           | 452.22 Cr   | 36.94 Cr       | 48.92 Cr       | 85.86 Cr         |
| 49 |                | Advanced Commerce University     | 445.75 Cr   | 36.62 Cr       | 48.64 Cr       | 85.26 Cr         |
| 14 |                | Indian Technology University     | 455.77 Cr   | 35.94 Cr       | 48.56 Cr       | 84.5 Cr          |
| 48 |                | Central Commerce University      | 442.78 Cr   | 35.18 Cr       | 48.51 Cr       | 83.69 Cr         |
| 5  |                | Regional Law University          | 440.85 Cr   | 33.14 Cr       | 48.43 Cr       | 81.57 Cr         |

## STEP 4D — Minimum Data Threshold Gate

```

WITH default_stats AS (
  SELECT
    i.institution_id,
    i.institution_name,
    i.institution_type,
    COUNT(*) AS loan_count,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
    ROUND(default_count * 100.0 / loan_count, 2) AS default_rate
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name, i.institution_type
  HAVING COUNT(l.loan_id) >= 50
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  institution_type AS `Institution Type`,
  CONCAT(default_rate, '%') AS `Default Rate (%)`,
  CASE
    WHEN default_rate > 15 THEN 'Blacklist'
    WHEN default_rate > 12 THEN 'Monitor'
    ELSE 'Expand'
  END AS `Risk Classification`
FROM default_stats
ORDER BY default_rate DESC;

```

| Institution ID | Institution Name                 | Institution Type | Default Rate (%) | Risk Classification |
|----------------|----------------------------------|------------------|------------------|---------------------|
| 29             | State Commerce Institute         | Institute        | 20.87%           | Blacklist           |
| 33             | Central Engineering Institute    | Institute        | 20.75%           | Blacklist           |
| 34             | Modern Management Institute      | Institute        | 20.57%           | Blacklist           |
| 31             | Regional Commerce University     | University       | 20.56%           | Blacklist           |
| 30             | National Engineering Institute   | Institute        | 20.40%           | Blacklist           |
| 32             | National Commerce Institute      | Institute        | 20.39%           | Blacklist           |
| 50             | Government Medical College       | College          | 20.11%           | Blacklist           |
| 28             | Regional Medical University      | University       | 17.14%           | Blacklist           |
| 25             | Central Commerce College         | College          | 16.86%           | Blacklist           |
| 26             | State Arts & Science University  | University       | 16.69%           | Blacklist           |
| 24             | Modern Law College               | College          | 16.57%           | Blacklist           |
| 27             | Central Management College       | College          | 16.44%           | Blacklist           |
| 23             | Advanced Technology University   | University       | 16.32%           | Blacklist           |
| 36             | Indian Arts & Science University | University       | 13.38%           | Monitor             |
| 49             | Advanced Commerce University     | University       | 13.21%           | Monitor             |
| 8              | State Engineering Institute      | Institute        | 13.18%           | Monitor             |
| 13             | State Medical University         | University       | 12.96%           | Monitor             |
| 38             | Central Management College       | College          | 12.88%           | Monitor             |
| 12             | Central Commerce University      | University       | 12.80%           | Monitor             |
| 21             | Central Law University           | University       | 12.79%           | Monitor             |
| 46             | Regional Law College             | College          | 12.68%           | Monitor             |
| 7              | Indian Law University            | University       | 12.67%           | Monitor             |
| 48             | Central Commerce University      | University       | 12.65%           | Monitor             |
| 4              | Advanced Arts & Science College  | College          | 12.61%           | Monitor             |
| 2              | Central Law University           | University       | 12.61%           | Monitor             |

## STEP 4E — Partnership Review Dashboard (Final Output)

```

WITH institution_summary AS (
  SELECT
    i.institution_id,
    i.institution_name,
    i.institution_type,
    COUNT(*) AS loan_count,
    COUNT(DISTINCT l.customer_id) AS customers,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2) AS default_rate,
    ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2) AS default_amount
  FROM institutions i
  INNER JOIN loans l
    ON i.institution_id = l.institution_id
  GROUP BY i.institution_id, i.institution_name, i.institution_type
  HAVING COUNT(l.loan_id) >= 50
),
multiplier_data AS (
  SELECT
    *,
    CASE
      WHEN loan_count >= 200 THEN 1.5
      WHEN loan_count >= 50 THEN 1.0
      ELSE 0.5
    END AS volume_multiplier
  FROM institution_summary
)
SELECT
  institution_id AS `Institution ID`,
  institution_name AS `Institution Name`,
  institution_type AS `Institution Type`,
  loan_count AS `Loans`,
  customers AS `Customers`,
  CONCAT(default_amount, ' Cr') AS `Default Amount`,
  CONCAT(default_rate, '%') AS `Default Rate (%)`,
  ROUND((1 - default_rate / 100) * volume_multiplier * 100, 2) AS `Health Score`,
  CASE
    WHEN default_rate > 15 THEN 'Blacklist'
    WHEN default_rate > 8 THEN 'Monitor'

```

```

        ELSE 'Expand'
    END AS `Partner Flag`
FROM multiplier_data
ORDER BY default_rate DESC;

```

| Institution ID | Institution Name                 | Institution Type | Loans | Customers | Default Amount | Default Rate (%) | Health Score | Partner Flag |
|----------------|----------------------------------|------------------|-------|-----------|----------------|------------------|--------------|--------------|
| 29             | State Commerce Institute         | Institute        | 10005 | 9501      | 83.96 Cr       | 20.87%           | 118.70       | Blacklist    |
| 33             | Central Engineering Institute    | Institute        | 9878  | 9420      | 83.5 Cr        | 20.75%           | 118.88       | Blacklist    |
| 34             | Modern Management Institute      | Institute        | 10165 | 9660      | 88.54 Cr       | 20.57%           | 119.15       | Blacklist    |
| 31             | Regional Commerce University     | University       | 9915  | 9433      | 85.99 Cr       | 20.56%           | 119.16       | Blacklist    |
| 30             | National Engineering Institute   | Institute        | 9889  | 9466      | 85.15 Cr       | 20.40%           | 119.40       | Blacklist    |
| 32             | National Commerce Institute      | Institute        | 9947  | 9489      | 82.81 Cr       | 20.39%           | 119.42       | Blacklist    |
| 50             | Government Medical College       | College          | 9780  | 9326      | 79.94 Cr       | 20.11%           | 119.84       | Blacklist    |
| 28             | Regional Medical University      | University       | 10134 | 9617      | 68.86 Cr       | 17.14%           | 124.29       | Blacklist    |
| 25             | Central Commerce College         | College          | 10009 | 9527      | 66.67 Cr       | 16.86%           | 124.71       | Blacklist    |
| 26             | State Arts & Science University  | University       | 10108 | 9640      | 68.38 Cr       | 16.69%           | 124.97       | Blacklist    |
| 24             | Modern Law College               | College          | 10151 | 9642      | 64.85 Cr       | 16.57%           | 125.15       | Blacklist    |
| 27             | Central Management College       | College          | 9984  | 9497      | 64.64 Cr       | 16.44%           | 125.34       | Blacklist    |
| 23             | Advanced Technology University   | University       | 10121 | 9602      | 63.76 Cr       | 16.32%           | 125.52       | Blacklist    |
| 36             | Indian Arts & Science University | University       | 9897  | 9443      | 50.59 Cr       | 13.38%           | 129.93       | Monitor      |
| 49             | Advanced Commerce University     | University       | 9926  | 9436      | 48.64 Cr       | 13.21%           | 130.19       | Monitor      |
| 8              | State Engineering Institute      | Institute        | 9860  | 9387      | 47.59 Cr       | 13.18%           | 130.23       | Monitor      |
| 13             | State Medical University         | University       | 10259 | 9725      | 51.87 Cr       | 12.96%           | 130.56       | Monitor      |
| 38             | Central Management College       | College          | 10039 | 9525      | 50.51 Cr       | 12.88%           | 130.68       | Monitor      |
| 12             | Central Commerce University      | University       | 10082 | 9592      | 51.42 Cr       | 12.80%           | 130.80       | Monitor      |
| 21             | Central Law University           | University       | 9956  | 9487      | 49.09 Cr       | 12.79%           | 130.82       | Monitor      |
| 46             | Regional Law College             | College          | 9886  | 9385      | 47.21 Cr       | 12.68%           | 130.98       | Monitor      |
| 7              | Indian Law University            | University       | 10024 | 9549      | 49.63 Cr       | 12.67%           | 131.00       | Monitor      |
| 48             | Central Commerce University      | University       | 9916  | 9443      | 48.51 Cr       | 12.65%           | 131.03       | Monitor      |
| 4              | Advanced Arts & Science College  | College          | 10166 | 9654      | 49.19 Cr       | 12.61%           | 131.09       | Monitor      |
| 2              | Central Law University           | University       | 10107 | 9593      | 48.92 Cr       | 12.61%           | 131.09       | Monitor      |

## Custom Column

```

CASE
    WHEN default_rate >= 1.5 * (SELECT AVG(default_rate) FROM institution_summary)
    THEN 'Suspend'
    WHEN default_rate >= (SELECT AVG(default_rate) FROM institution_summary) THEN
    'Review'
    WHEN default_rate >= 0.5 * (SELECT AVG(default_rate) FROM institution_summary)
    THEN 'Reward'
    ELSE 'Retain'
END AS partner_flag

CASE WHEN EXISTS (
    SELECT 1 FROM customers c
    JOIN loans l
    ON c.customer_id = l.customer_id
    WHERE l.institution_id = institution_summary.institution_id AND c.cibil_score >
    900
)
THEN 'Data Quality Review Required'
ELSE 'Data Clean' END AS anomaly_flag

```

A fixed threshold like `default_rate > 15%` assumes you already know what "bad" looks like before seeing the data, but that number is arbitrary and breaks down across different economic conditions, loan types, or time periods. Comparing against the portfolio average is self-calibrating: if the whole portfolio is performing poorly (say average is 18%), a 15% institution is actually your best partner, not a suspension candidate. It also means your flags automatically adapt as the portfolio changes — no one needs to manually update hardcoded thresholds every quarter.

## WHY Gate

### 1. The institution with 0 defaults — what action did you assign it, and why? Write your classification AND your reasoning. ("I assigned it X because Y, not Z because A.")

'National Law Institute' (Institution ID 17) is the only institution with 0 defaults. I classified the institution as 'Reward' and not 'Retain', 'Review' OR 'Suspend', because its default rate of 0.00% falls below 0.5x the portfolio average, making it the strongest performer/institution in the entire dataset. I did not classify it as 'Retain' because Retain is reserved for institutions between 0.5x and 1x the average, which still carry moderate risk. 'Review' and 'Suspend' are ruled out entirely since those require a default rate at or above the portfolio average.

### 2. Write your Suspension Justification for the top-ranked institution to be suspended. Format it as if going to a legal review: Institution name, default rate, loan count, total defaulted exposure, which threshold was exceeded, recommended action, and what monitoring must happen post-suspension.

- Institution: State Commerce Institute (Institution ID: 29) - Default Rate: 20.87% - Total Loans Disbursed: 10005 - Total Defaulted Exposure: 83.96 Crores - Threshold Exceeded: 20.87% exceeds the 1.5x portfolio average threshold - Recommended Action: Immediate suspension of new loan disbursements pending full portfolio audit. No new student loan applications to be processed through this institution until review is complete. - Post-Suspension Monitoring Requirements: - 83.96 Cr in defaulted exposure must be assigned to collections with a 90-day recovery report due to the Partnership Director - Institution returned Data Clean on the anomaly check, meaning no CIBIL >900 irregularities were detected; however a manual data audit is still recommended given the volume - Partnership agreement to be placed under legal hold; no contract renewals or amendments permitted during suspension period.

### 3. If you were the Partnership Director and received your final output — what is the FIRST call you make? Who do you call, about which institution, and what specifically do you say in that call?

If I were the partnership director that receive my own final output, then I would first try to call the institution who had 0% default rate despite having 10026 loans. This result did not come from a rounding error, but due to the structural difference of how you guys are operating compared to the other institutions. Before I spend the next few weeks on suspension hearings for the bottom of this portfolio, I want to understand what you're doing differently like, what's your screening process, your student selection criteria, your collections touchpoints, whatever it is. Because if we can document it and replicate it, that's worth more to this organization than suspending 10 bad partners.

